

Lab 3, Act 2

Counting detections: binomial or Poisson?

NRES 746 – name: _____ group: _____

Picking up where we left off

In Act 1 you dealt with *uncertainty about a single flagged image*. Now set that aside – assume genetic testing has confirmed the target carnivore is genuinely present at a particular site, so there’s no more classifier doubt to worry about. The new question is about *detection*: even when the animal is really there, a camera doesn’t catch it every single night. Suppose, based on many past deployments at similar sites, biologists estimate that on any given night there’s a $p = 0.3$ probability the camera captures at least one image of the animal, independent from night to night.

Before you do anything else: close your laptop and put it away. Part 1 and the group check-in that follows are pen-and-paper only.

Part 1: Binomial by hand (individual, ~15 min)

A camera is deployed for $N = 8$ nights.

1a. Why is “number of nights with a detection, out of 8” a reasonable binomial random variable here? Name the assumptions the binomial distribution requires, and say briefly why each one seems plausible (or not) for this camera.

1b. Using the binomial pmf, $P(X = k) = \binom{N}{k} p^k (1 - p)^{N-k}$, compute by hand the probability of exactly 3 detections in the 8 nights ($N = 8$, $k = 3$, $p = 0.3$). Show your work.

Group check-in: compare your hand calculation (group, ~5-8 min)

Compare your answer to 1b with your group and agree on a single, correct value before moving on.

Laptops open now. Verify your hand calculation and go further.

Part 2: Verify, visualize, and compare to Poisson (laptops, ~20-25 min)

2a. Use `dbinom()` to check your hand-computed answer from 1b. Do they match?

2b. Plot the full distribution of detections across all 9 possible outcomes ($k = 0$ to 8) with `barplot(dbinom(0:8, 8, 0.3))`. Which number of detections is most likely?

2c. A colleague suggests it would be simpler to just model detections with a Poisson distribution instead, using the same mean, $\lambda = 8 \times 0.3 = 2.4$. Plot `barplot(dpois(0:8, 2.4))` right next to your binomial plot from 2b (e.g. with `par(mfrow=c(1,2))`). Do the two distributions look like reasonable substitutes for one another?

2d. Compute both variances – binomial: $Np(1-p)$; Poisson: λ – and record them below. Does the mismatch (or lack of one) explain what you saw in 2c?

Binomial variance = _____ Poisson variance = _____

2e. Now suppose instead the camera ran for $N = 100$ nights, but the per-night detection probability was much lower, $p = 0.024$ (same mean, $100 \times 0.024 = 2.4$). Repeat 2c and 2d for this new N and p (`dbinom(0:8, 100, 0.024)` vs. `dpois(0:8, 2.4)`). What changed?

2f. Based on 2c-2e, finish this rule of thumb in your own words: “A *Poisson distribution is a good stand-in for a binomial distribution when...*”

2g. Back in the original 8-night scenario ($p = 0.3$): name one realistic reason the constant, independent per-night detection probability assumption from 1a might be violated in the field (think about weather, the animal’s behavior, or the camera itself).

Group discussion and revision (~5 min)

Compare your plots, variances, and rule of thumb from 2f with your group.

Feel free to revise any of your answers above based on the discussion (it’s fine to cross out and rewrite – I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion (“muddiest points”) or topics you want to review or dust off going forward.
